

various protein and gene ontology databases.

The complexity of life processes

Living organisms are examples of complex systems, functioning by means of complex pathways which in turn may also interact with each other. In other words, living organisms are not merely a sum of their genes; the interaction between genes gives rise to many emergent properties, which cannot be studied only by summing up the functions of individual genes.

Wiki Pathways ([wikipathways.org](http://www.wikipathways.org)) is a community resource for curating cell pathways. It is an open resource for users and contributors alike.

Cytoscape (www.cytoscape.org) is an open source software platform, specifically designed to visualise complex biological networks. It also offers Web based interactive views of public bioscience databases. Indeed, the best of network and exploratory data analysis tools need to be exploited to analyse the complexity of cell processes.

Data mining

Data is exploding at a rate far greater than that at which humans can assimilate or make sense of it. Data mining and machine learning techniques help researchers with decision making, classification and clustering of biological data, and also to understand complex relationships and find patterns in biological data.

Orange (orange.biolab.si) is an open source tool available for data mining through visual programming or Python scripting. It also has extra features for text mining, bioinformatics and data analytics. Other machine learning software tools such as *scikit-learn* can be used equally well to mine biological data.

Biological text mining works on literature produced in the areas of medical and molecular biology. The technique helps in the extraction of information for biological processes and diseases. For example, information mined from biology literature has helped to determine the connection between magnesium and migraine headaches. Interestingly, this fact was discovered by Don R. Swanson, an American information scientist and was clinically validated only later.

Epidemiological models to study the spread of diseases

Mathematical models are especially useful to study and manage infectious diseases. Such models help to predict the degree of vulnerability of different areas, the rate of spread of the disease and also the measures that need to be taken to control the spread of the disease. Mathematical models and software help authorities calculate the effectiveness of public medical interventions like mass vaccination programs.

The R statistical computing environment has a set of Outbreak Tools under its R-epi project. These tools help make sense of the diverse and complex outbreak data by statistically analysing and visualising it. SAGES (Suite for Automated

Global Electronic bioSurveillance) is a collection of open source software tools that can be used to closely examine the spread of any disease.

Biosimulation

Nobody expected that *in vivo* and *in vitro* would be followed by *in silico*. *In silico* is another way of saying that the organ or organism is 'living' inside a computer or is being simulated by it. Some specialised companies market virtual organs or simulated disease models.

In silico biosimulation of diseases is helping scientists to model and understand them like never before. These models not only help simulate the disease but also help create virtual populations which are susceptible to the disease. The simulation of complex biological networks and pathways helps in the processes of drug discovery and testing. It enables researchers to take informed decisions, thus optimising clinical trials to check the effectiveness of new drugs.

PySB is a Python framework for modelling, simulating and visualising biochemical processes. BioPython is a set of Python tools for computations related to biology. Finally (for this article, at least), BioLinux is a Linux distribution for biologists and life science researchers. Based on Ubuntu 14.04 LTS, it offers a large collection of software packages for the life sciences. The open source model is also encouraging new forms of distributed research and crowd sourcing, one of them being the Open Source Drug Discovery project (osdd.net).

The use of computer science to deal with challenges in the life sciences is a huge leap for research in this area as it has opened up many new opportunities. It has also introduced new techniques that make it easier to understand the complexity of living systems. Moreover, it also helps us to do away with many tedious procedures that need to be performed repeatedly in a wet-lab. Undoubtedly, the new computational approaches used are transforming the life sciences for the better and additionally, making research more open, collaborative and accessible. 

References

- [1] <http://www.technologyreview.com/graphiti/427720/bases-to-bytes/>
- [2] <https://en.wikipedia.org/wiki/Genome>
- [3] <http://www.ebi.ac.uk/about/background>
- [4] http://en.wikipedia.org/wiki/Literature-based_discovery
- [5] <http://www.r-project.org/>
- [6] <https://sites.google.com/site/thereproject/r-pac/about>
- [7] <http://www.jhuapl.edu/sages/>
- [8] http://biopython.org/wiki/Main_Page
- [9] <http://environmentalomics.org/bio-linux/>
- [10] <http://opensource.com/health/11/11/open-source-cancer-research>

By: Apurva Talgery

The author is an avid user of open source software. Her areas of interest include systems biology and bioinformatics.